Seokbin Yoon

Ph.D. Student | George Washington University | Washington, DC, USA

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Education

George Washington University Aug. 2026 - Present
Doctor of Philosophy | *Mechanical and Aerospace Engineering*
- Advisor: Dr. Peng Wei

Korea Aerospace University Mar. 2022 - Feb. 2024
Master of Science | *Air Transport, Transportation, and Logistics*
- Dissertation: Trajectory Reconstruction and Prediction for Vectored Area Navigation Arrivals
- Advisor: Dr. Keumjin Lee

Korea Aerospace University Mar. 2020 - Feb. 2022
Bachelor of Science | *Air Transport, Transportation, and Logistics*

Publications

Peer-Reviewed Journal Articles:

[11] H. Kim, B. Kang, **Seokbin Yoon**, & K. Lee (2026). On the Feasibility of Image-Based Representations for Air Traffic Complexity Estimation.
Journal of the Air Transport Research Society. [Paper]

[10] **Seokbin Yoon** & K. Lee (2026). Multi-Agent Inverted Transformer for Flight Trajectory Prediction.
IEEE Transactions on Intelligent Transportation Systems. [Paper]

[9] **Seokbin Yoon** & K. Lee (2025). Aircraft Trajectory Dataset Augmentation in Latent Space.
International Journal of Aeronautical & Space Sciences. [Paper]

[8] **Seokbin Yoon** & K. Lee (2025). Aircraft Trajectory Prediction with Inverted Transformer.
IEEE Access. [Paper] [Code]

Conference Proceedings:

[7] J. Lee, **Seokbin Yoon**, K. Lee, H. Baik, & S. Jung (2026). Airport Terminal Passenger Queue Forecasting for Departure Gates and Security Checkpoints.
AIAA/IEEE Digital Avionics Systems Conference (DASC). [Paper] [Code]

[6] H. Kim, B. Kang, **Seokbin Yoon**, D. Jeon, & K. Lee (2026). Reading the ATC Display: Vision Transformers Learn Air Traffic Complexity from Radar Imagery.
Air Transport Research Society World Conference.

[5] K. Kim, **Seokbin Yoon**, & K. Lee (2026). Probabilistic Multi-Agent Aircraft Landing Time Prediction.
AIAA SciTech Forum. [Paper] [Code]

[4] H. Chai, **Seokbin Yoon**, & K. Lee (2025). Learning to Explain Air Traffic Situation.
International Air Transportation Research & Development Symposium. [Paper]

[3] J. Kang, **Seokbin Yoon**, K. Lee, & S. H. Kim (2024). Urban Air Mobility Fleet Rebalancing with Real-Time Updates on Estimated Time of Arrival.
AIAA/IEEE Digital Avionics Systems Conference (DASC). [Paper]

[2] **Seokbin Yoon** & K. Lee (2023). Improving Aircraft Trajectory Prediction Accuracy with Over-Sampling Technique.
AIAA/IEEE Digital Avionics Systems Conference (DASC). [Paper]

[1] N. H. Gray, S. Shin, **Seokbin Yoon**, Y. Park, H. Chai, K. Lee, H. Kim, & S. Jung (2023). Safety and Capacity Analysis Framework for Integrated UAM Operation in Airports.
AIAA/IEEE Digital Avionics Systems Conference (DASC). [Paper]

Manuscripts Under Review:

[4] B. Liu, D. Vu, **Seokbin Yoon**, & P. Wei. Benchmarking General-Purpose and Fine-Tuned Speech Recognition Models for Air Traffic Control
(under review)

[3] K. Kim, **Seokbin Yoon**, P. Wei, T. Kwon, & K. Lee. Intrinsic Terminal Airspace Capacity Estimation via Generative Air Traffic Modeling
(under review)

[2] **Seokbin Yoon**, K. Kim, T. Kwon, K. Lee, A. Zongo, & P. Wei. DiTSG++: Separation-Guided Multi-Agent Diffusion Transformers for Conditional Air Traffic Scenario Generation.
(under review)

[1] **Seokbin Yoon**, K. Kim, T. Kwon, K. Lee, & P. Wei. DiTSG: Context-Aware Air Traffic Scene Generation with Multi-Agent Diffusion Transformers.
(under review) [Preprint] [Project] [Code]

Academic and Volunteer Service

Journal Reviewer

- AIAA Journal of Aerospace Information Systems
- IEEE Internet of Things Journal
- Journal of Open Aviation Science

Conference Reviewer

- IEEE International Conference on Robotics and Automation

Honors and Awards

Best Paper of the Session, AIAA/IEEE Digital Avionics Systems Conference	2026
Best Student Paper, AIAA SciTech Forum	2026
Best Paper of the Session, AIAA/IEEE Digital Avionics Systems Conference	2024
Best Paper of the Session, AIAA/IEEE Digital Avionics Systems Conference	2023
President's Award, Korea Aerospace University	2023
Brain Korea 21 Plus (BK21+) Research Scholarship, Korean Government	2022

Skills

Programming: Python, Julia, MATLAB, LaTeX
Frameworks & Libraries: PyTorch, NumPy, Scikit-learn, OpenCV

Licenses & Certifications

Air Traffic Controller 2023
Ministry of Land, Infrastructure and Transport, South Korea

Machine Learning with Python: From Linear Models to Deep Learning 2023
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